

Upay SecurePay - AI-Powered MFS Transaction Protection

Upay SecurePay is an intelligent security layer for mobile financial services that helps prevent wrong-number transfers, scams, fraud, and account takeover before money is fully transferred.

Its main features are:

- **Verified Recipient Name Preview:** When a user enters a mobile number, Upay retrieves the verified recipient name stored in its KYC database and shows it before the transaction is confirmed. This helps users immediately verify that they are sending money to the correct person.
- **Unknown Recipient Detection:** Checks whether the receiver is a previous recipient or a completely new user.
- **30-Second Undo:** Risky or first-time transfers remain pending for 30 seconds, allowing the sender to cancel them before settlement.
- **AI Risk Score:** Every transaction is analyzed using amount, recipient history, device, time, login behavior, transaction frequency, and spending pattern.
- **Behavioral Anomaly Detection:** AI learns the user's normal activity and detects unusual or suspicious behavior.
- **Recipient Trust Score:** Recipients are classified as trusted, recognized, unknown, or suspicious based on transaction history and explainable risk signals.
- **Fraud Network Detection:** AI identifies suspicious accounts, repeated scam patterns, unusual money flows, and connected fraud networks.
- **Adaptive Security:** Low-risk payments remain fast, while higher-risk transactions receive additional verification, a short delay, or a temporary review hold instead of an unexplained automatic rejection.
- **15-Minute Secure Lock:** Possible account takeover or highly suspicious activity can trigger a temporary outgoing hold that requires secure re-verification or authorized review before money can move.
- **Explainable Security Alerts:** Users are clearly told why an activity was flagged and what protection was applied.
- **Continuous Learning:** Confirmed fraud, cancelled transactions, and user feedback help improve future fraud detection.

Core idea:

Show users who they are paying before they send, keep normal payments fast, and intelligently slow down suspicious transactions or require extra verification when AI detects risk.

Upay AI Financial Coach

Upay AI Financial Coach is a personal money-management assistant that helps users understand their spending, save better, improve financial habits, and make smarter financial decisions using Upay transaction patterns. **For the hackathon prototype, all transaction data will be synthetic or simulated rather than real customer data.**

Main features:

- **AI Spending Analysis:** Automatically categorizes expenses and shows where the user's money is going.
- **"Why Did I Run Out of Money?" Report:** Explains the main reasons behind low balance or overspending.
- **Personalized Budget Plan:** Creates weekly or monthly budgets based on income and spending habits.
- **Savings Goal Planner:** Users set a target and deadline, and AI creates a step-by-step saving plan.
- **Smart Saving Suggestions:** Recommends how much the user can realistically save.
- **Spending Habit Coach:** Detects repeated or avoidable spending patterns and suggests practical improvements without forcing a financial decision.
- **AI Financial Assistant:** Users can ask questions about their own finances and receive personalized, explainable answers.
- **Expense Forecasting:** Predicts upcoming regular expenses and warns users before they may run short of money.
- **Overspending Alerts:** Warns users when they are spending faster than usual or exceeding their budget.
- **Financial Health Score:** Provides a simple informational score based on spending, saving, budgeting, and financial stability. It is not used to approve or deny credit.
- **Weekly/Monthly Money Report:** Summarizes income, expenses, savings progress, and areas for improvement.
- **Personalized Financial Learning:** Gives short financial lessons based on the user's actual behavior.

- **AI Smart Discount Alerts:** If a user frequently shops at a particular merchant and that merchant publishes a discount or offer through Upay, AI identifies whether the offer is relevant to the user's habits and, with user opt-in, notifies them before their next likely purchase. The goal is useful savings, not encouraging unnecessary spending.

Core idea:

Turn Upay into an AI financial companion that helps users spend wisely, save consistently, reach financial goals, and discover useful offers based on their financial behavior.

Upay Family Pool - Shared Expense & Contribution Manager

Upay Family Pool is a group-based money management feature where families, roommates, friends, or small teams can create a shared financial group, assign expenses, collect contributions, and track who has paid what.

Main features:

- **Create a Pool:** One user creates a group and invites other Upay users.
- **Group Roles:** The creator becomes the manager and can assign admins or members.
- **Shared Expenses:** The manager can add expenses such as rent, groceries, bills, tuition, subscriptions, or family costs.
- **Automatic Split:** Expenses can be divided equally, by percentage, by custom amount, or only among selected members.
- **Assigned Payments:** Each member can clearly see how much they need to pay and the deadline.
- **One-Tap Contribution:** Members pay their share directly through Upay. In the hackathon prototype, this payment flow can be simulated.
- **Payment Tracking:** The group can see paid, unpaid, partially paid, and overdue contributions.
- **Reminders:** Members receive automatic reminders before and after due dates.
- **Shared Goal Fund:** Groups can also create savings goals for trips, emergencies, education, events, or major purchases.
- **Group Spending History:** Shows total collected, total spent, contribution history, and individual member records.

AI Integration

- **AI Expense Split Assistant:** Suggests fair split options based on past contributions or user-defined rules. Members approve the final split before it is applied.
- **AI Budget Planner:** Helps the group create a monthly budget based on recurring expenses.
- **AI Contribution Prediction:** Predicts upcoming shared expenses and recommends how much each member should prepare.
- **AI Late-Payment Risk:** Identifies contributions that may be missed based on schedule and payment patterns, then sends earlier reminders. It does not penalize or restrict members.
- **AI Group Spending Insights:** Explains where the group is spending most and suggests ways to reduce unnecessary costs.
- **AI Savings Planner:** Creates a contribution plan for shared goals based on target amount, deadline, and members' declared contribution preferences.
- **AI Financial Assistant:** Members can ask questions such as "How much do I owe this month?", "Why did our family expenses increase?", or "Can we reach our Eid travel goal in three months?"

Core idea

Upay Family Pool turns informal family and group money collection into a transparent, organized, and AI-assisted shared finance system.

It solves the common problem of repeatedly asking "**Who paid?**", "**How much is left?**", and "**Who still owes money?**" while making shared financial planning much easier.

Hackathon prototype approach

Data: Use synthetic Upay-style transaction, recipient, merchant, and group-expense data only; no real customer PII is required. **AI role:** risk detection, anomaly detection, forecasting, recommendation, and explanation. **Validation:** measure wrong-transfer prevention, fraud-detection quality and false alerts, budget/expense forecast accuracy, savings-goal progress, and shared-payment completion. **Responsible AI:** important security or financial actions stay explainable and require appropriate user confirmation or review.